

3D POINT CLOUD RECONSTRUCTION

from Monocular Video Data

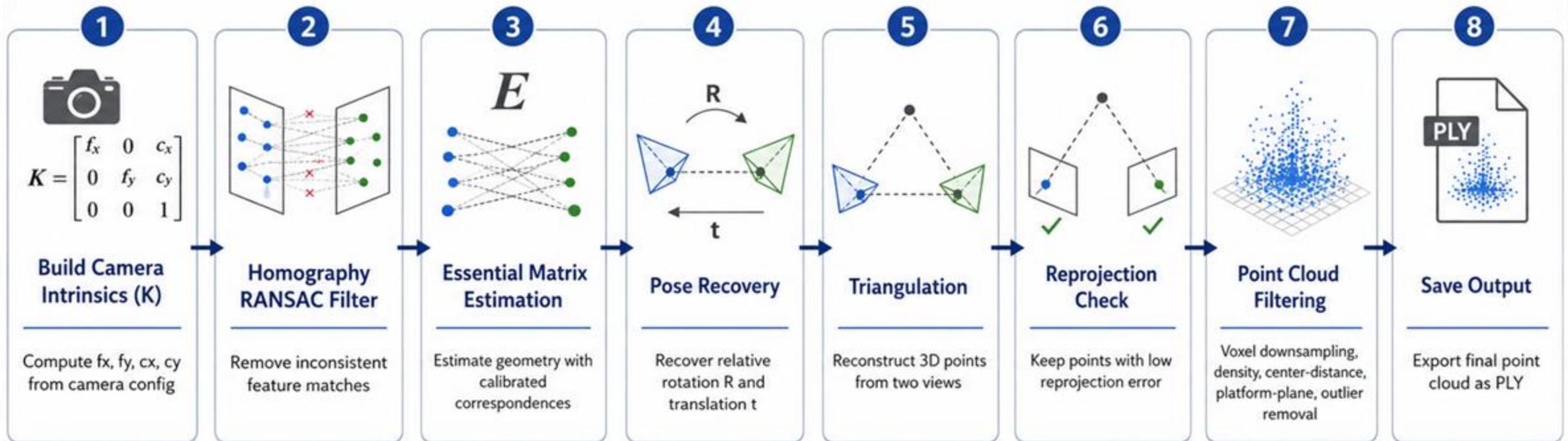
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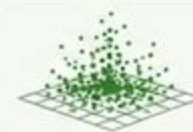
OUTLINE

1. New Pipeline
2. Pipeline Implementation Status
3. Deep Dive:
 - Feature Extraction
 - Camera Pose & Triangulation
4. Image to 3D Object
5. Conclusion

Pipeline



Input: matched feature points from two frames



Output: filtered 3D point cloud

DEEP DIVE: FEATURE EXTRACTION

SuperPoint + LightGlue · Deep Learning Feature Detection & Matching



SuperPoint — Keypoint Detector

- ▶ Self-supervised deep learning (VGG encoder + 2 decoder heads)
- ▶ Detector head → heatmap of interest points
- ▶ Descriptor head → 256-D dense feature map per point
- ▶ ~300 keypoints per frame · repeatable & rotation-robust
- ▶ Homographic Adaptation training — no manual labels needed

LightGlue — Feature Matcher

- ▶ Transformer with self- + cross-attention between keypoint sets
- ▶ Adaptive pruning — discards unreliable keypoints early
- ▶ Early-exit mechanism — stops when confidence is sufficient
- ▶ Faster than LoFTR & more accurate than SIFT-based RANSAC

Method Comparison

Method	Speed	Accuracy	GPU	Fixed-cam
SIFT (classical)	Med	Low	No	✗
LoFTR / RoMa	Slow	High	Yes	⚠ Overkill
Depth Anything v2	Slow	Med	Yes	✗
SuperPoint + LightGlue ✓	Fast	High	Yes	✓

⚠ Compute Cost Consideration

SuperPoint + LightGlue require CUDA GPU. Depth Anything v2 rejected — GPU cost prohibitive for batch.

~300

keypoints
/frame

256-D

descriptor
vectors

100%

success rate
78 configs

DEEP DIVE: CAMERA POSE & TRIANGULATION

We do NOT estimate camera pose — we know it from the orbit rig geometry

Key insight: Pose is known from the orbit rig

Rig parameters from dataset README:

- `H = 410 mm height` • `R = 400 mm radius`
- `inclination ≈ 45°`
- Camera always looks at platform centre (origin)
- Constant angular velocity around the orbit

Camera position for orbit angle θ (Y-up world frame):

$$\mathbf{C}(\theta) = [R \cdot \cos \theta, \quad H, \quad R \cdot \sin \theta]$$

$$\theta \text{ for frame } i = (i \cdot \text{stride} / \text{total_frames}) \cdot 360^\circ$$

Avoids classical SfM problems entirely:

- No scale ambiguity — sparse cloud directly in millimetres
- No pose drift • no bundle adjustment for metric scale

Two triangulation paths

- Pair-by-pair DLT (default)
 - Triangulate each matched (f_0, f_0+8) pair independently
 - `cv2.triangulatePoints` with $K[R|t]$ from `orbit_pose()`
 - Cheirality check: drop points behind either camera
- HLoc multi-view tracks (`--use-hloc`)
 - Union-Find merges chains: $K \leftrightarrow L \leftrightarrow M \rightarrow$ one track
 - Triangulate with widest baseline $\rightarrow$ lower depth uncertainty
 - Each 3D point unique, no near-duplicates

K matrix from physics

Nikon D780 sensor 35.9×24.0 mm, square pixels.
 $fx = \text{focal_length_mm} / (\text{sensor_width} / \text{image_width})$

No factory calibration in dataset — focal length defaults to 100 mm, refined to ~147 mm via iterative scale-hint feedback (best fit to GT diagonals).

LIMITATIONS

Focal-length calibration variance

Scale factor spread 0.31 – 1.53. Assumes a single focal length across the dataset, but per-part effective focal drifts slightly.

→ *Bounds absolute-mm accuracy at ~15 – 25 mm Chamfer*

FUNDAMENTAL

Platform-plane filter hurts F-Score

Cleaner clouds visually, but F @ 5 % drops ~0.02 vs pre-filter.

→ *Ground truth includes some near-platform points; removing them looks like missed geometry*

TRADE-OFF

Depth densification is compute-heavy

Depth Anything v2 is ~10× slower per frame than sparse reconstruction.

→ *Only 1 sample fully dense so far · batch dense run pending*

ONGOING

Approaches evaluated but not adopted

Reprojection-error filtering, higher keypoint counts (>4096), match-gap variations, denser stride, deeper depth models.

→ *Each either broke the pipeline or gave marginal metric gains at large runtime cost*

EXPLORED

RESULT

- The results overall are consistent
- Median scale factor: 0.91

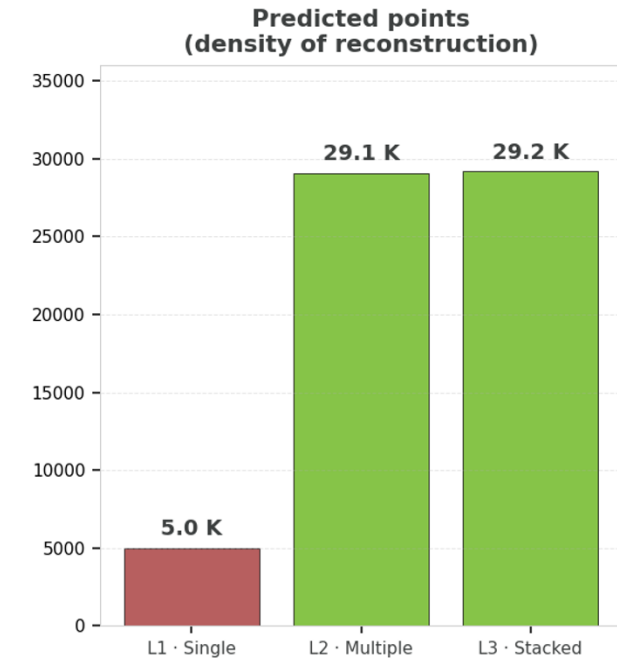
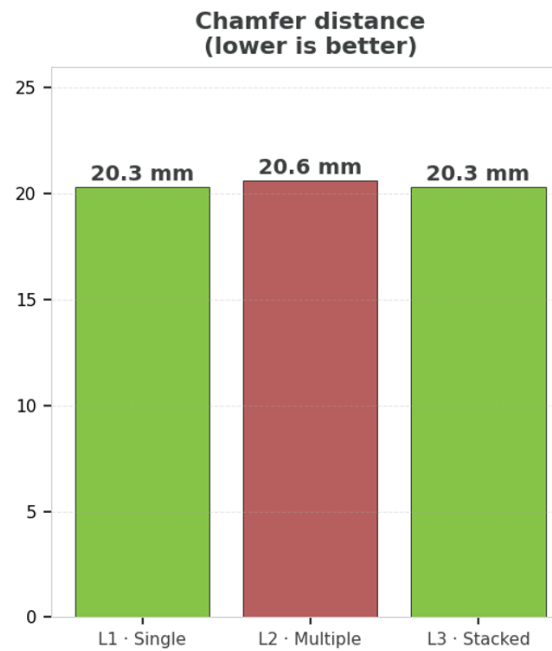
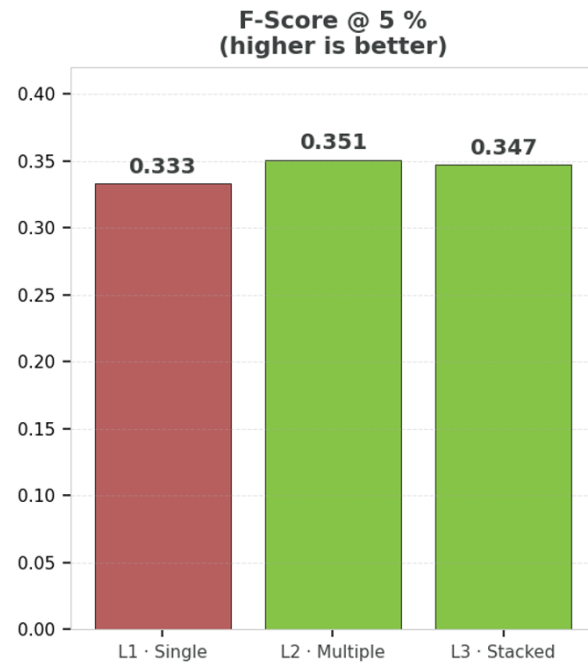
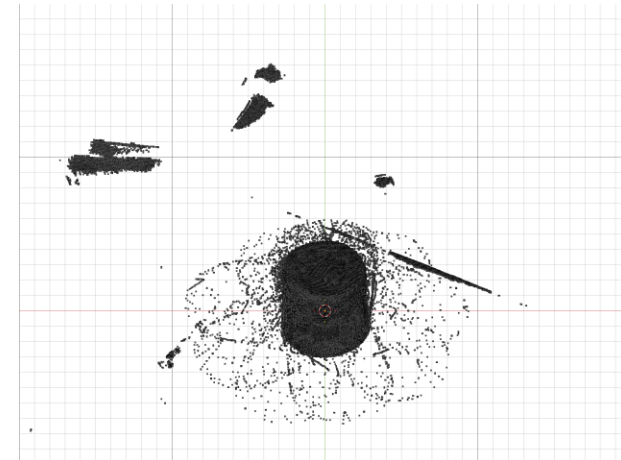
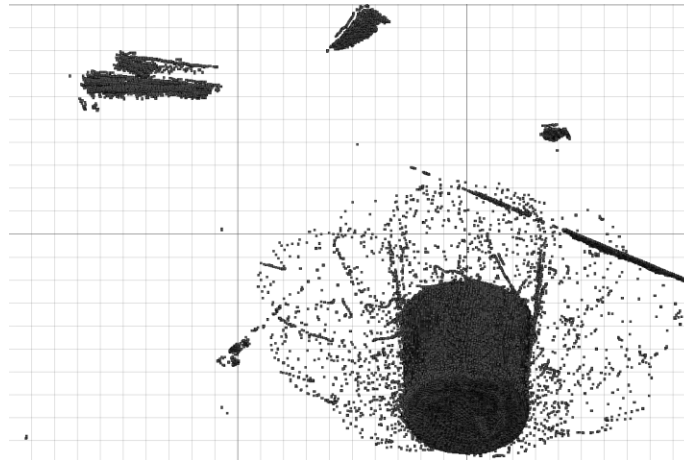
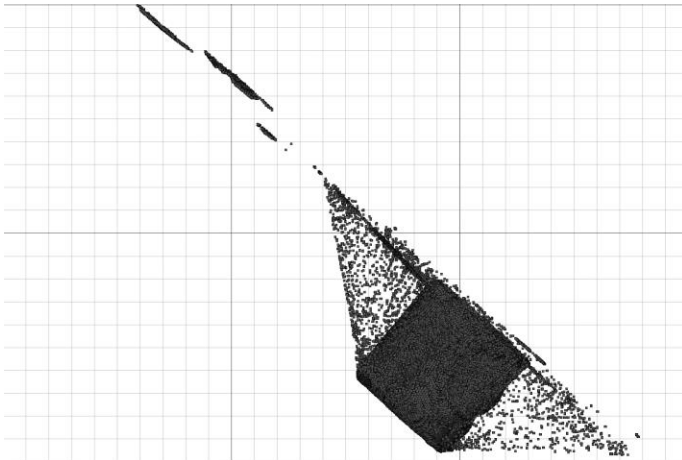


IMAGE TO 3D OBJECT

- Model: DUS t 3R
 - Used checkpoint: DUS t 3R_ViTLarge_BaseDecoder_512_dpt
- It is fast and looks promising, but geometrically poor
 - No information about camera poses and intrinsics
- Results shown in x, y, z order



IMPROVEMENTS

- Improvements for feature extraction and matching
 - Fewer data points are extracted compared to ground truth
 - Single: 0.36%
 - Multiple: 2.23%
 - Stacked: 2.53%
- Filtering out points known not to belong to the object

CONCLUSION

- Pipeline is fast but not practical
 - Reconstruction takes around 10 seconds (RTX 5080)
- Accuracy too low for real-world use
 - Reconstructed cloud is noisy
 - Contaminations, point estimations
- LiDAR sensor is better suited for high accuracy

FUTURE WORKS

- Geometrically-Grounded Generative 3D
 - DUS3R can be used as baseline
- Fast and calibration-free, but arbitrary scale
- No knowledge of object or platform
- **Goal:** A generative model that produces geometrically correct objects, rather than just fast but approximate



THANK YOU